Cloud, Microservices and Applications

Lab Assignment 2

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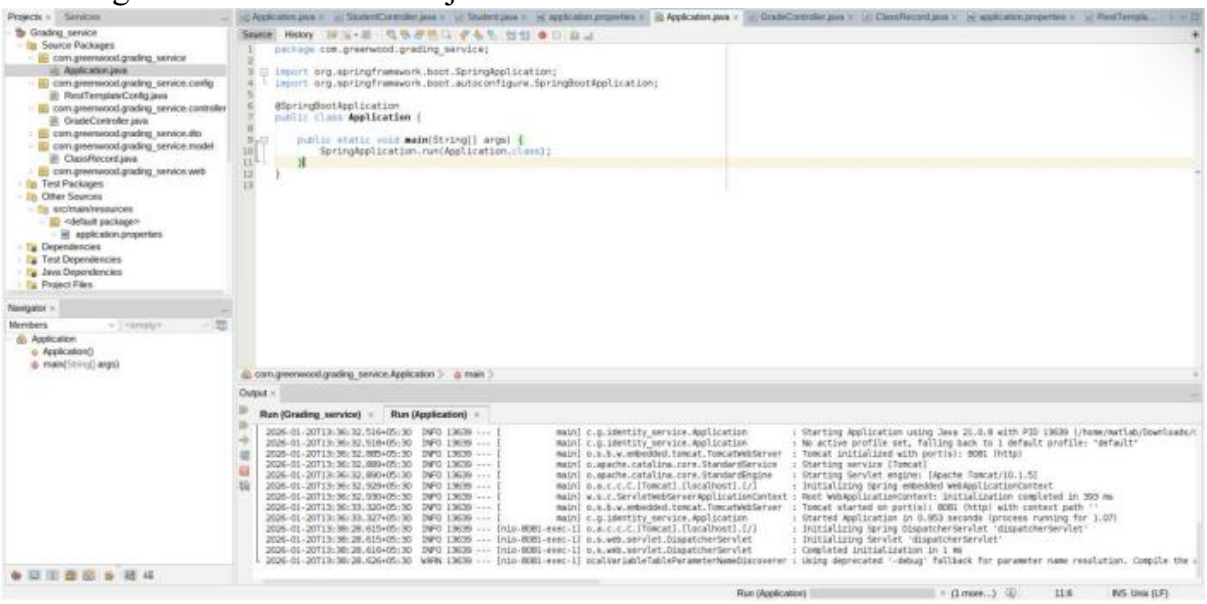
Running the microservice on Docker

1) Bundling the microservice as a .jar file.



2) Create a docker file.

3) Place the docker file in the microservice folder
4) Create a container in docker

```
matlab@sjt516scope019:~$ docker image ls
                                                    i Info →   U  In Use
IMAGE    ID             DISK USAGE   CONTENT SIZE   EXTRA
matlab@sjt516scope019:~$ docker run eclipse -temurin
Unable to find image 'eclipse:latest' locally
docker: Error response from daemon: pull access denied for eclipse, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
matlab@sjt516scope019:~$ docker run eclipse -temurin:17-jdk
Unable to find image 'eclipse:latest' locally
docker: Error response from daemon: pull access denied for eclipse, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
matlab@sjt516scope019:~$ docker run eclipse-temurin:17-jdk
Unable to find image 'eclipse-temurin:17-jdk' locally
17-jdk: Pulling from library/eclipse-temurin
f12be9d0c3db: Pull complete
69a71bf0bc7a: Pull complete
6086452543Bd: Pull complete
2eb378d50f5b: Pull complete
a3629ac5b9f4: Pull complete
fc38ec6cd66d: Download complete
a165f5665a9f: Download complete
Digest: sha256:330029ffc8a75f4c74ed35c882378e1854a6f7ebdc53c43f25b587dba0fd2f93
Status: Downloaded newer image for eclipse-temurin:17-jdk
Feb 10, 2026 7:18:18 AM java.util.prefs.FileSystemPreferences$1 run
INFO: Created user preferences directory.
|  Welcome to JShell -- Version 17.0.18
|  For an introduction type: /help intro
```

```
jshell> matlab@sjt516scope019:~$ docker image ls
                                                    i Info →   U  In Use
IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
eclipse-temurin:17-jdk   330029ffc8a7       671MB          210MB    U
matlab@sjt516scope019:~$ docker container ls
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
matlab@sjt516scope019:~$ docker run eclipse-temurin:17-jdk
Feb 10, 2026 7:21:28 AM java.util.prefs.FileSystemPreferences$1 run
INFO: Created user preferences directory.
|  Welcome to JShell -- Version 17.0.18
|  For an introduction type: /help intro

jshell> matlab@sjt516scope019:~$ docker container ls
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
matlab@sjt516scope019:~$ docker create eclipse-temurin:17-jdk
36e1ef36e253e35834da732cc37facef357bcd3ca704d3a3c95e0716a4e0406d
matlab@sjt516scope019:~$ docker container ls
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
matlab@sjt516scope019:~$ docker run -i -t  eclipse-temurin:17-jdk
Feb 10, 2026 7:26:00 AM java.util.prefs.FileSystemPreferences$1 run
INFO: Created user preferences directory.
|  Welcome to JShell -- Version 17.0.18
|  For an introduction type: /help intro

jshell>
```

```
matlab@sjt516scope019:~$ docker container ls
CONTAINER ID   IMAGE                    COMMAND                  CREATED          STATUS          PORTS     NAMES
99c1c41ed8b0   eclipse-temurin:17-jdk   "/__cacert_entrypoin…"   22 seconds ago   Up 22 seconds             angry_merkle
matlab@sjt516scope019:~$
```

5) Check if the microservice is working also

```
matlab@sjt516scope019:~$ java -jar /home/matlab/NetBeansProjects/grading-service/target/grading-service-1.0-SNAPSHOT.jar

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 :: Spring Boot ::                (v3.0.2)

2026-02-10T13:08:30.875+05:30  INFO 11587 --- [           main] c.greenwood.grading.service.Application  : Starting Application using Java 21.0.8 with PID 11587 (/home/matlab/NetBeansP
rojects/grading-service/target/grading-service-1.0-SNAPSHOT.jar started by matlab in /home/matlab)
2026-02-10T13:08:30.876+05:30  INFO 11587 --- [           main] c.greenwood.grading.service.Application  : No active profile set, falling back to 1 default profile: "default"
2026-02-10T13:08:31.312+05:30  INFO 11587 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port(s): 8082 (http)
2026-02-10T13:08:31.317+05:30  INFO 11587 --- [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-02-10T13:08:31.317+05:30  INFO 11587 --- [           main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/10.1.5]
2026-02-10T13:08:31.354+05:30  INFO 11587 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContext
2026-02-10T13:08:31.355+05:30  INFO 11587 --- [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 453 ms
2026-02-10T13:08:31.796+05:30  INFO 11587 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port(s): 8082 (http) with context path ''
2026-02-10T13:08:31.803+05:30  INFO 11587 --- [           main] c.greenwood.grading.service.Application  : Started Application in 1.089 seconds (process running for 1.266)
```

6) Build the image

```
matlab@sjt516scope019:~/NetBeansProjects$ docker build -t grading-service:latest ./grading-service
[+] Building 0.8s (8/8) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 157B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                                                     0.0s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [1/3] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:330029ffc8a75f4c74ed35c882378e1054a6f7ebdc53c43f25b587dba0fd2f93  0.1s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:330029ffc8a75f4c74ed35c882378e1054a6f7ebdc53c43f25b587dba0fd2f93  0.0s
 => [internal] load build context                                                                                             0.1s
 => => transferring context: 24.47MB                                                                                          0.1s
 => [2/3] WORKDIR /app                                                                                                        0.0s
 => [3/3] COPY ./target/*.jar ./app.jar                                                                                       0.0s
 => exporting to image                                                                                                        0.4s
 => => exporting layers                                                                                                       0.4s
 => => exporting manifest sha256:34599d03f5a8f22482a3fe0b45ff19a9f972fbf03fee63937fc4a7e1c2418589                             0.0s
 => => exporting config sha256:e4e0d3694943c2f1c04b204043865c7fd8cccb574a321f2a7eb7920a36eb6cbf                               0.0s
 => => exporting attestation manifest sha256:60dce70fa0bb5d34de925dc2b633ca61f0b9681a2c0d6b3c67deef80658af5a0                 0.0s
 => => exporting manifest list sha256:9920c732cc2e76719f4c6df6614f47b64cfcc55c497ebcaa4e60fae7cbe7dc48                        0.0s
 => => naming to docker.io/library/grading-service:latest                                                                     0.0s
 => => unpacking to docker.io/library/grading-service:latest                                                                  0.1s
```

7) Run the microservice on docker

```
matlab@sjt516scope019:~/NetBeansProjects$ docker run grading-service

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 :: Spring Boot ::                (v3.0.2)

2026-02-10T08:04:36.448Z  INFO 1 --- [           main] c.greenwood.grading.service.Application  : Starting Application using Java 17.0.18 with PID 1 (/app/app.jar started by root in /a
pp)
2026-02-10T08:04:36.449Z  INFO 1 --- [           main] c.greenwood.grading.service.Application  : No active profile set, falling back to 1 default profile: "default"
2026-02-10T08:04:36.860Z  INFO 1 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat initialized with port(s): 8082 (http)
2026-02-10T08:04:36.866Z  INFO 1 --- [           main] o.apache.catalina.core.StandardService   : Starting service [Tomcat]
2026-02-10T08:04:36.867Z  INFO 1 --- [           main] o.apache.catalina.core.StandardEngine    : Starting Servlet engine: [Apache Tomcat/10.1.5]
2026-02-10T08:04:36.902Z  INFO 1 --- [           main] o.a.c.c.C.[Tomcat].[localhost].[/]       : Initializing Spring embedded WebApplicationContext
2026-02-10T08:04:36.903Z  INFO 1 --- [           main] w.s.c.ServletWebServerApplicationContext : Root WebApplicationContext: initialization completed in 430 ms
2026-02-10T08:04:37.315Z  INFO 1 --- [           main] o.s.b.w.embedded.tomcat.TomcatWebServer  : Tomcat started on port(s): 8082 (http) with context path ''
2026-02-10T08:04:37.325Z  INFO 1 --- [           main] c.greenwood.grading.service.Application  : Started Application in 1.032 seconds (process running for 1.191)
```